Fractions Made Easy

A Quick-Start Guide to Mastering Fractions

Target Audience: Grades 5–8

Lazard Legacy Math Academy

Author: Gregory Lazard Jr.

"Dreams can come true. Better late than never."

Table of Contents

Section 1: What is a Fraction?

A **fraction** represents a part of a whole. It is written as one number over another, separated by a fraction bar (or line).

Parts of a Fraction:
• Numerator (top number): Tells how many parts you HAVE.
• Denominator (bottom number): Tells how many EQUAL parts the whole is divided into.

Types of Fractions:

• **Proper Fraction:** Numerator is LESS than the denominator.

Example: 3/4, 2/5, 7/8

• **Improper Fraction:** Numerator is GREATER than or equal to the denominator.

Example: 7/4, 5/3, 11/6

• **Mixed Number:** A whole number combined with a proper fraction.

Example: 1 3/4, 2 1/2, 3 2/5

Equivalent Fractions:

Fractions that look different but represent the SAME amount are called equivalent fractions. You can create equivalent fractions by multiplying or dividing both the numerator and denominator by the same number.

```
Example: 1/2 = 2/4 = 3/6 = 4/8 = 5/10
```

Simplifying Fractions:

To simplify (or reduce) a fraction, divide both the numerator and denominator by their Greatest Common Factor (GCF).

```
Example: Simplify 12/18
```

GCF of 12 and 18 = 6
$12 \div 6 = 2$, $18 \div 6 = 3$
Answer: 2/3

Comparing Fractions:
To compare fractions with different denominators, find a common denominator or cross-multiply. The fraction with the larger cross-product on top is larger.

```
Example: Compare 3/4 and 5/7
```

$3 \times 7 = 21$ vs $5 \times 4 = 20$
$21 > 20$, so $3/4 > 5/7$

Section 2: Adding & Subtracting Fractions

Adding/Subtracting with SAME Denominator:
Add (or subtract) the numerators. Keep the denominator the same. Simplify if possible.

Adding/Subtracting with DIFFERENT Denominators:
Step 1: Find the LCD (Least Common Denominator).
Step 2: Rewrite each fraction with the LCD.
Step 3: Add (or subtract) the numerators.
Step 4: Keep the LCD as the denominator.
Step 5: Simplify if possible.

Worked Examples — Adding Fractions:

Example 1: 2/7 + 3/7

Same denominator!
Add numerators: 2 + 3 = 5
Keep denominator: 7
Answer: 5/7

Example 2: 1/3 + 1/4

Different denominators: LCD of 3 and 4 = 12
1/3 = 4/12 (multiply top and bottom by 4)
1/4 = 3/12 (multiply top and bottom by 3)
4/12 + 3/12 = 7/12
Answer: 7/12

Example 3: 5/6 + 2/9

LCD of 6 and 9 = 18

5/6 = 15/18
2/9 = 4/18
15/18 + 4/18 = 19/18 = 1 1/18
Answer: 1 1/18

Example 4: 3/4 + 5/8

LCD of 4 and 8 = 8
3/4 = 6/8
6/8 + 5/8 = 11/8 = 1 3/8
Answer: 1 3/8

Worked Examples — Subtracting Fractions:

Example 5: 7/8 – 1/8

Same denominator!
Subtract numerators: 7 – 1 = 6
6/8 → simplify by GCF(6,8)=2 → 3/4
Answer: 3/4

Example 6: 5/6 – 1/4

LCD of 6 and 4 = 12
5/6 = 10/12
1/4 = 3/12
10/12 – 3/12 = 7/12
Answer: 7/12

Example 7: 4/5 – 2/3

LCD of 5 and 3 = 15
4/5 = 12/15
2/3 = 10/15
12/15 – 10/15 = 2/15
Answer: 2/15

Example 8: 11/12 – 1/2

LCD of 12 and 2 = 12
1/2 = 6/12
11/12 – 6/12 = 5/12
Answer: 5/12

Common Mistakes to Avoid:

✗ Do NOT add the denominators! (1/3 + 1/4 ≠ 2/7)

✗ Do NOT forget to simplify your answer.

✗ Do NOT forget to convert improper fractions to mixed numbers when needed.

Section 3: Multiplying & Dividing Fractions

Multiplying Fractions:
Step 1: Multiply the numerators (top × top).
Step 2: Multiply the denominators (bottom × bottom).
Step 3: Simplify the result.

Tip: You can cross-cancel BEFORE multiplying to make the numbers smaller!

Dividing Fractions — "Keep, Change, Flip":
Step 1: KEEP the first fraction exactly as it is.
Step 2: CHANGE the division sign (÷) to multiplication (×).
Step 3: FLIP the second fraction (write its reciprocal).
Step 4: Multiply and simplify.

Worked Examples — Multiplying:

Example 1: 2/3 × 4/5

Numerators: 2 × 4 = 8
Denominators: 3 × 5 = 15
No common factors to simplify
Answer: 8/15

Example 2: 3/4 × 2/9

Cross-cancel: 3 and 9 share factor 3 → becomes 1/4 × 2/3
Numerators: 1 × 2 = 2
Denominators: 4 × 3 = 12
Simplify: 2/12 = 1/6
Answer: 1/6

Example 3: 5/7 × 7/10

Cross-cancel: 7 cancels; 5 and 10 share 5
Becomes: 1/1 × 1/2
Answer: 1/2

Example 4: 1/2 × 8/9

Cross-cancel: 2 and 8 share 2 → becomes 1/1 × 4/9
= 4/9
Answer: 4/9

Worked Examples — Dividing:

Example 5: 3/4 ÷ 1/2

Keep 3/4
Change ÷ to ×
Flip 1/2 → 2/1
3/4 × 2/1 = 6/4
Simplify: 6/4 = 3/2 = 1 1/2
Answer: 1 1/2

Example 6: 5/6 ÷ 2/3

Keep 5/6
Flip 2/3 → 3/2
5/6 × 3/2 = 15/12
Simplify: 15/12 = 5/4 = 1 1/4
Answer: 1 1/4

Example 7: 7/8 ÷ 7/4

Keep 7/8
Flip 7/4 → 4/7
7/8 × 4/7 = 28/56
Simplify: 28/56 = 1/2
Answer: 1/2

Example 8: 2/9 ÷ 4/3

Keep 2/9
Flip 4/3 → 3/4
2/9 × 3/4 = 6/36
Simplify: 6/36 = 1/6
Answer: 1/6

Section 4: Mixed Numbers & Improper Fractions

Converting Improper Fraction → Mixed Number:
Divide the numerator by the denominator.
The quotient is the whole number.
The remainder becomes the new numerator (keep the same denominator).

Converting Mixed Number → Improper Fraction:
Multiply the whole number by the denominator.
Add the numerator to that result.
Write this sum over the original denominator.

Worked Examples:

Example 1: Convert 11/4 to a mixed number

Divide: 11 ÷ 4 = 2 remainder 3
Whole number = 2
Remainder = 3 (keep denominator 4)
Answer: 2 3/4

Example 2: Convert 3 2/5 to an improper fraction

Multiply whole number by denominator: 3 × 5 = 15
Add numerator: 15 + 2 = 17
Keep denominator: 5
Answer: 17/5

Example 3: Convert 22/7 to a mixed number

Divide: 22 ÷ 7 = 3 remainder 1

Whole number = 3, remainder = 1

Answer: 3 1/7

Section 4 (continued): Operations with Mixed Numbers

To add, subtract, multiply, or divide mixed numbers:
Convert to improper fractions first, then perform the operation, then convert back to a mixed number if needed.

Example 4: Add: 2 1/3 + 1 2/3

Convert: 2 1/3 = 7/3
Convert: 1 2/3 = 5/3
Same denominator: 7/3 + 5/3 = 12/3
Simplify: 12/3 = 4
Answer: 4

Example 5: Multiply: 1 1/2 × 2 2/3

Convert: 1 1/2 = 3/2
Convert: 2 2/3 = 8/3
Multiply: 3/2 × 8/3 = 24/6
Simplify: 24/6 = 4
Answer: 4

Quick Reference — Converting:

Improper Fraction	Mixed Number
5/3	1 2/3
7/4	1 3/4
11/5	2 1/5
9/2	4 1/2
13/6	2 1/6

17/8	2 1/8

Section 5: Practice Problems

40 Fraction Problems — Show All Work

1) 1/5 + 2/5 =

2) 3/8 + 1/4 =

3) 2/3 + 5/6 =

4) 7/10 + 1/5 =

5) 1/2 + 3/7 =

6) 5/6 – 1/6 =

7) 3/4 – 1/3 =

8) 7/8 – 1/2 =

9) 11/12 – 3/4 =

10) 4/5 – 1/10 =

Practice Problems (continued)

11) $2/3 \times 3/5 =$

12) $4/7 \times 7/8 =$

13) $5/9 \times 3/10 =$

14) $1/4 \times 8/9 =$

15) $6/11 \times 11/12 =$

16) $1/2 \div 1/4 =$

17) $3/5 \div 6/10 =$

18) $7/8 \div 3/4 =$

19) $2/9 \div 1/3 =$

20) $5/6 \div 5/12 =$

Practice Problems (continued)

21) Convert 15/4 to a mixed number.

22) Convert 2 5/8 to an improper fraction.

23) Convert 19/6 to a mixed number.

24) 1 1/2 + 2 3/4 =

25) 3 1/3 – 1 2/3 =

26) Simplify 18/24.

27) Simplify 35/42.

28) 3/4 + 2/3 + 1/12 =

29) (2/5) × (5/6) × (3/4) =

30) 4/5 ÷ 2 =

Practice Problems (continued)

31) Which is greater: 5/8 or 7/12?

32) Find a fraction between 1/3 and 1/2.

33) 2 1/4 × 1 1/3 =

34) 5/6 – 2/9 =

35) 7/15 + 2/5 =

36) 3/10 ÷ 9/20 =

37) Convert 4 3/7 to an improper fraction.

38) Simplify 48/60.

39) 1/2 + 1/3 + 1/6 =

40) A pizza has 8 slices. You eat 3. What fraction remains?

Section 6: Answer Key

1) 3/5
2) 5/8
3) 3/2 = 1 1/2
4) 9/10
5) 13/14
6) 4/6 = 2/3
7) 5/12
8) 3/8
9) 1/6
10) 7/10
11) 2/5
12) 1/2
13) 1/6
14) 2/9
15) 1/2
16) 2
17) 1
18) 7/6 = 1 1/6
19) 2/3
20) 2

Answer Key (continued)

21) 3 3/4
22) 21/8
23) 3 1/6
24) 4 1/4
25) 1 2/3
26) 3/4
27) 5/6
28) 18/12 = 3/2 = 1 1/2
29) 1/4
30) 2/5
31) 5/8 > 7/12 (compare: 5×12=60 vs 7×8=56)
32) 2/5 (answers vary)
33) 3
34) 11/18
35) 13/15
36) 2/3
37) 31/7
38) 4/5
39) 1
40) 5/8